

15.3 Transition Matrix



The core skills you'll master in this section:

- ▶ Each element of the transition matrix represents a transition probability.
- ▶ Each column of the transition matrix sums to 1.
- ▶ Multiplying a power of the transition matrix by the state vector derives the weather state after any number of days.
- ▶ After many weather transitions, the state vector converges to a fixed probability distribution, called the steady state.
- ▶ The (normalized) eigenvector of the transition matrix corresponding to eigenvalue 1 corresponds to the steady state.

Three Weather States

As shown in Figure 1, the weather in a certain region has only three states — sunny, cloudy, and rainy.

The next day's weather state depends only on the current weather:

- ◀ If today is sunny, tomorrow there's a 70% chance of sun, a 25% chance of clouds, and a 5% chance of rain.
- ◀ If today is cloudy, tomorrow there's a 45% chance of sun, a 30% chance it stays cloudy, and a 25% chance of rain.
- ◀ If today is rainy, tomorrow there's a 55% chance of sun, a 30% chance of clouds, and a 15% chance it stays rainy.

Looking at Figure 1, this is essentially a weighted directed graph.

Each node represents a weather state, and each directed edge represents a transition between weather states; the weight of a directed edge represents a probability value.

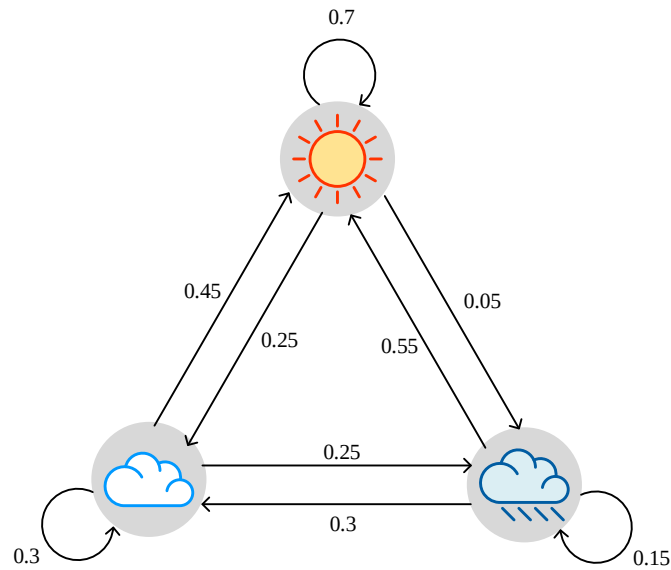


Figure 1. Transitions Among the Three Weather States

Adjacency Matrix

Based on the previous section, we can easily obtain Figure 1's adjacency matrix A

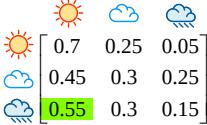
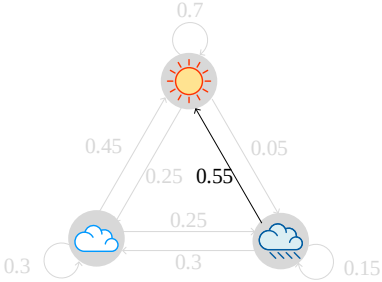
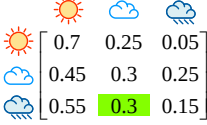
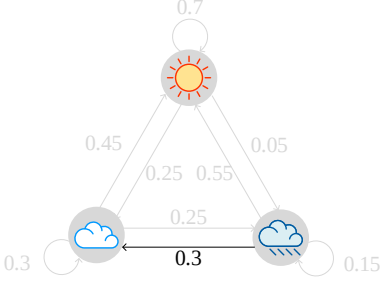
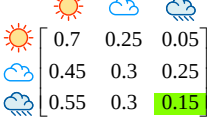
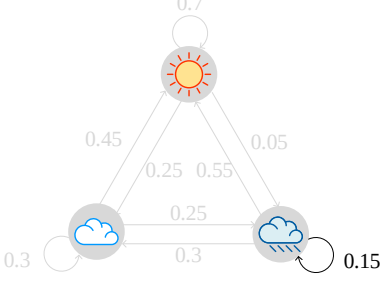
$$A = \begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix}$$

⚠ Note that each row of this adjacency matrix sums to 1; each row represents the probability distribution of jumping from the current node to other nodes.

Table 1. The Meaning of Each Element of Adjacency Matrix A

Transition	Element of Adjacency Matrix A	Directed Edge
Sunny → Sunny (70%)	$\begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix}$	

Sunny → Cloudy (25%)	$ \begin{matrix} & \text{Sunny} & \text{Cloudy} & \text{Rainy} \\ \text{Sunny} & 0.7 & 0.25 & 0.05 \\ \text{Cloudy} & 0.45 & 0.3 & 0.25 \\ \text{Rainy} & 0.55 & 0.3 & 0.15 \end{matrix} $	Diagram showing the transition from Sunny to Cloudy (25%). The Sunny node has a self-loop of 0.7 and transitions to Cloudy (0.25) and Rainy (0.05). The Cloudy node has a self-loop of 0.3 and transitions to Sunny (0.45) and Rainy (0.25). The Rainy node has a self-loop of 0.15 and transitions to Sunny (0.05) and Cloudy (0.3).
Sunny → Rainy (5%)	$ \begin{matrix} & \text{Sunny} & \text{Cloudy} & \text{Rainy} \\ \text{Sunny} & 0.7 & 0.25 & 0.05 \\ \text{Cloudy} & 0.45 & 0.3 & 0.25 \\ \text{Rainy} & 0.55 & 0.3 & 0.15 \end{matrix} $	Diagram showing the transition from Sunny to Rainy (5%). The Sunny node has a self-loop of 0.7 and transitions to Cloudy (0.25) and Rainy (0.05). The Cloudy node has a self-loop of 0.3 and transitions to Sunny (0.45) and Rainy (0.25). The Rainy node has a self-loop of 0.15 and transitions to Sunny (0.05) and Cloudy (0.3).
Cloudy → Sunny (45%)	$ \begin{matrix} & \text{Sunny} & \text{Cloudy} & \text{Rainy} \\ \text{Sunny} & 0.7 & 0.25 & 0.05 \\ \text{Cloudy} & 0.45 & 0.3 & 0.25 \\ \text{Rainy} & 0.55 & 0.3 & 0.15 \end{matrix} $	Diagram showing the transition from Cloudy to Sunny (45%). The Sunny node has a self-loop of 0.7 and transitions to Cloudy (0.25) and Rainy (0.05). The Cloudy node has a self-loop of 0.3 and transitions to Sunny (0.45) and Rainy (0.25). The Rainy node has a self-loop of 0.15 and transitions to Sunny (0.05) and Cloudy (0.3).
Cloudy → Cloudy (30%)	$ \begin{matrix} & \text{Sunny} & \text{Cloudy} & \text{Rainy} \\ \text{Sunny} & 0.7 & 0.25 & 0.05 \\ \text{Cloudy} & 0.45 & 0.3 & 0.25 \\ \text{Rainy} & 0.55 & 0.3 & 0.15 \end{matrix} $	Diagram showing the transition from Cloudy to Cloudy (30%). The Sunny node has a self-loop of 0.7 and transitions to Cloudy (0.25) and Rainy (0.05). The Cloudy node has a self-loop of 0.3 and transitions to Sunny (0.45) and Rainy (0.25). The Rainy node has a self-loop of 0.15 and transitions to Sunny (0.05) and Cloudy (0.3).
Cloudy → Rainy (25%)	$ \begin{matrix} & \text{Sunny} & \text{Cloudy} & \text{Rainy} \\ \text{Sunny} & 0.7 & 0.25 & 0.05 \\ \text{Cloudy} & 0.45 & 0.3 & 0.25 \\ \text{Rainy} & 0.55 & 0.3 & 0.15 \end{matrix} $	Diagram showing the transition from Cloudy to Rainy (25%). The Sunny node has a self-loop of 0.7 and transitions to Cloudy (0.25) and Rainy (0.05). The Cloudy node has a self-loop of 0.3 and transitions to Sunny (0.45) and Rainy (0.25). The Rainy node has a self-loop of 0.15 and transitions to Sunny (0.05) and Cloudy (0.3).

Rainy → Sunny (55%)	 $\begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix}$	
Rainy → Cloudy (30%)	 $\begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix}$	
Rainy → Rainy (15%)	 $\begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix}$	

Weather State Transitions

If the current weather is sunny, corresponding to the row vector $[1 \ 0 \ 0]$, the probabilities of each weather state on the next day can be obtained via the following matrix multiplication

$$[1 \ 0 \ 0] @ A = [1 \ 0 \ 0] @ \begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix} = [0.7 \ 0.25 \ 0.05]$$

The multiplication above is equivalent to extracting the first row of adjacency matrix A .

Similarly, if the current weather is cloudy, corresponding to $[0 \ 1 \ 0]$, the probabilities of each weather state on the next day can be obtained via the following matrix multiplication

$$[0 \ 1 \ 0] @ A = [0 \ 1 \ 0] @ \begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix} = [0.45 \ 0.3 \ 0.25]$$

Likewise, the equation below represents the probabilities of each weather state on the next day when the current weather is rainy $[0 \ 0 \ 1]$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} @ \mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.45 & 0.3 & 0.25 \\ 0.55 & 0.3 & 0.15 \end{bmatrix} = \begin{bmatrix} 0.55 & 0.3 & 0.15 \end{bmatrix}$$

Transition Matrix

The three equations above clearly aren't in the matrix-multiplication form we're used to; once transposed, they become the familiar form.

For example, after transposing, we get

$$\mathbf{A}^T @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.25 \\ 0.05 \end{bmatrix}$$

We give the adjacency matrix's transpose $\mathbf{A}^T$ a name – the transition matrix (transition matrix, left stochastic matrix, Markov matrix) $\mathbf{T}$, that is

$$\mathbf{T} = \mathbf{A}^T = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix}$$

This way, each column of transition matrix $\mathbf{T}$ sums to 1.

⚠ Note that, in the equation above, some references instead use the transpose (i.e., a matrix whose rows each sum to 1) as the transition matrix – this kind of transition matrix is called a right stochastic matrix; the state vector is a row vector, and the matrix multiplication requires a transpose.

We use the state vector $\mathbf{x}_t$ to represent the current weather, and $\mathbf{x}_{t+1}$ to represent the next day's weather.

⚠ Note that the elements of state vector $\mathbf{x}$ sum to 1.

As shown in Figure 2, when the current weather is sunny, the probabilities of each weather state on the next day can be written as the matrix multiplication

$$\mathbf{T} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.25 \\ 0.05 \end{bmatrix}$$

The equation above is equivalent to extracting the first column of transition matrix $\mathbf{T}$.

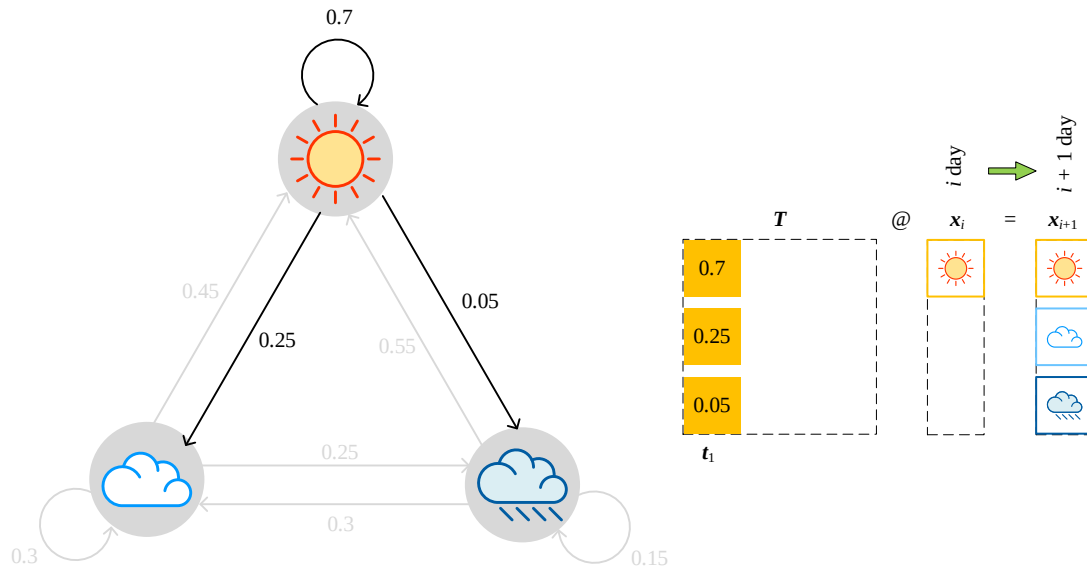


Figure 2. Transitioning From a Sunny Day to the Next Day's Weather State

As shown in Figure 3, when the current weather is cloudy, the probabilities of each weather state on the next day can be written as the matrix multiplication

$$T @ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.45 \\ 0.3 \\ 0.25 \end{bmatrix}$$

The equation above is equivalent to extracting the second column of transition matrix T .

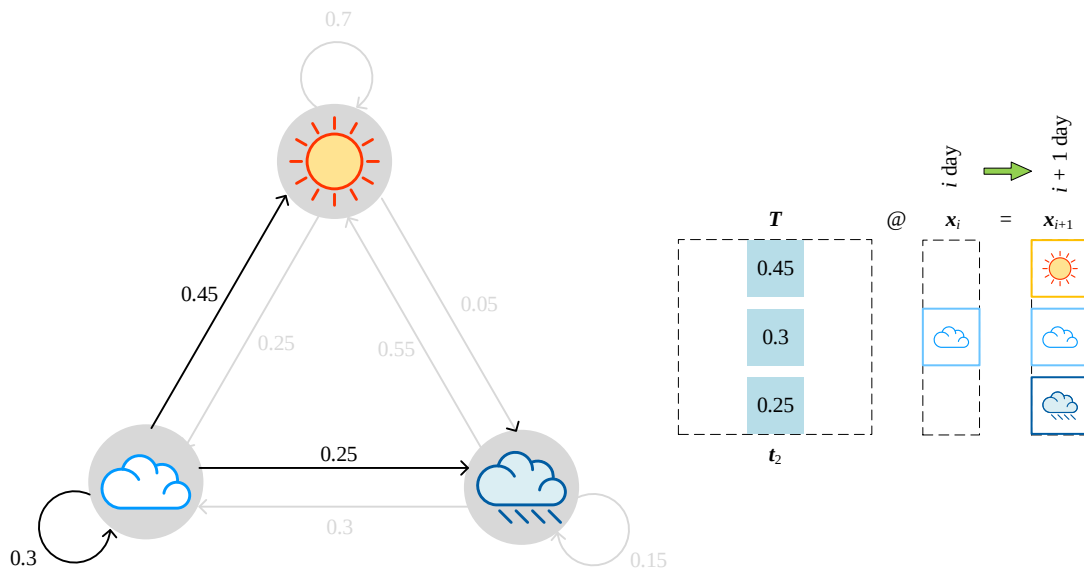


Figure 3. Transitioning From a Cloudy Day to the Next Day's Weather State

As shown in Figure 4, when the current weather is rainy, the probabilities of each weather state on the next day can be written as the matrix multiplication

$$T @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.55 \\ 0.3 \\ 0.15 \end{bmatrix}$$

The equation above is equivalent to extracting the third column of transition matrix T .

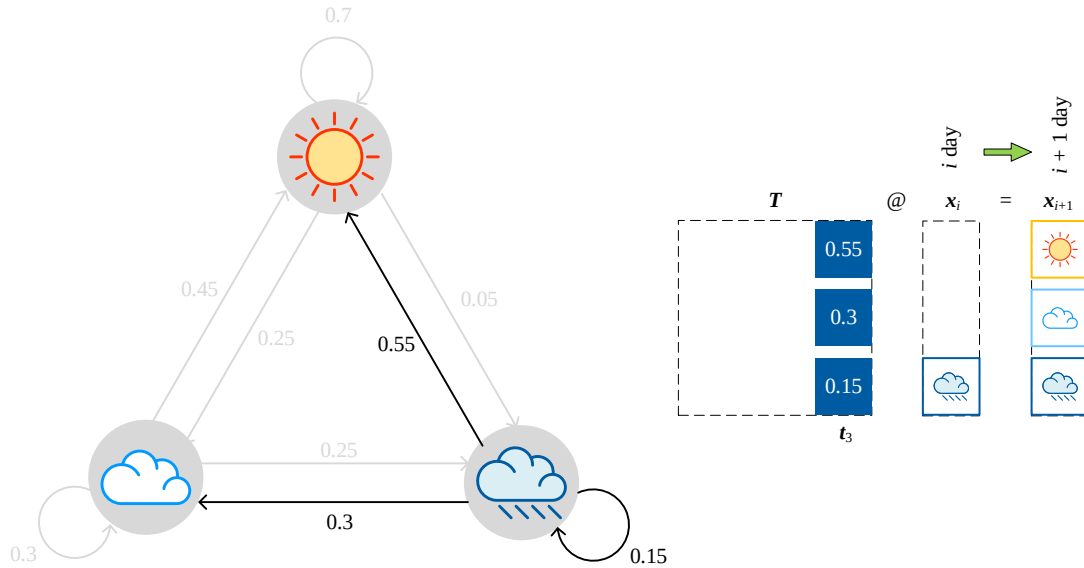


Figure 4. Transitioning From a Rainy Day to the Next Day's Weather State

As shown in Figure 5, the relationship among transition matrix T , the current weather state x_i , and the next day's weather state x_{i+1} is as follows

$$x_{i+1} = T x_i$$

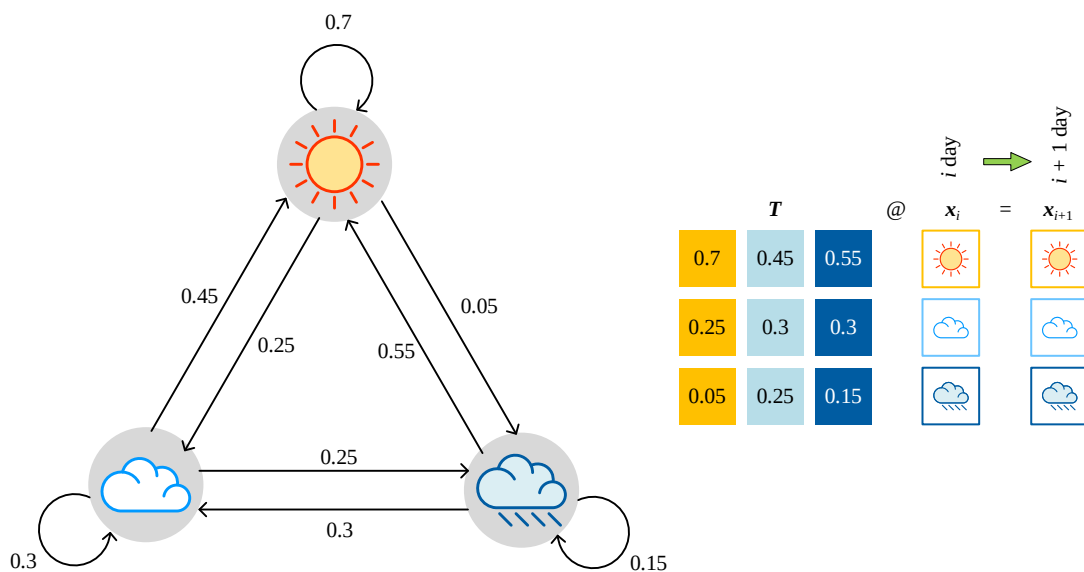


Figure 5. The Transition Matrix of the Weather-State Directed Graph

Transition Matrix Row Vectors

Looking at transition matrix T from the perspective of row vectors, we get the three figures Figure 6, Figure 7, and Figure 8.

Figure 6 shows the computation for the probability of transitioning from each of the three current weather states to a sunny next day.

Please analyze Figure 7 and Figure 8 on your own.

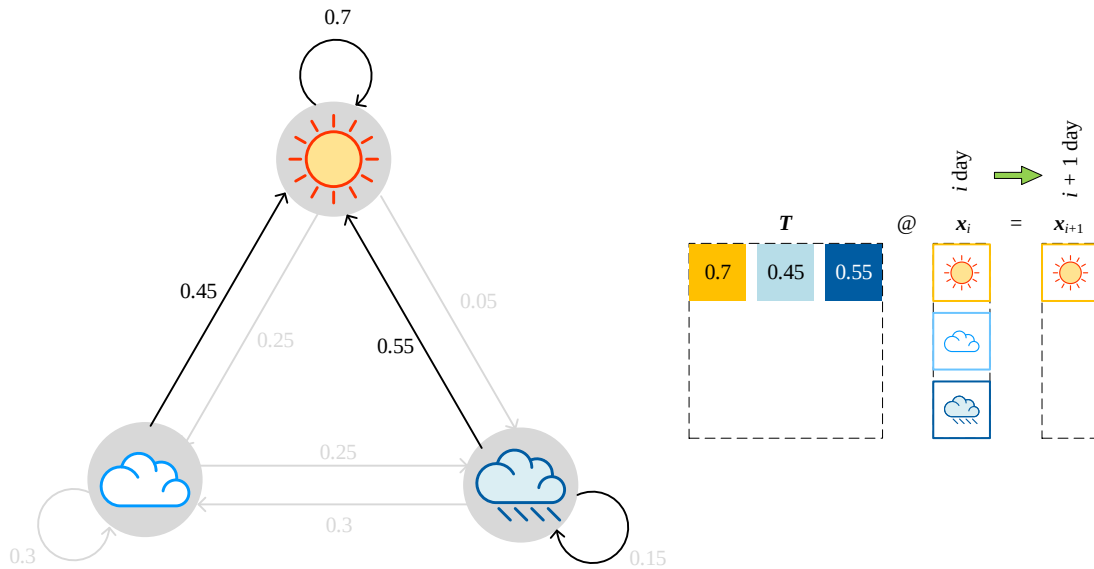


Figure 6. The Computation for Transitioning From the Three Current Weather States to a Sunny Next Day

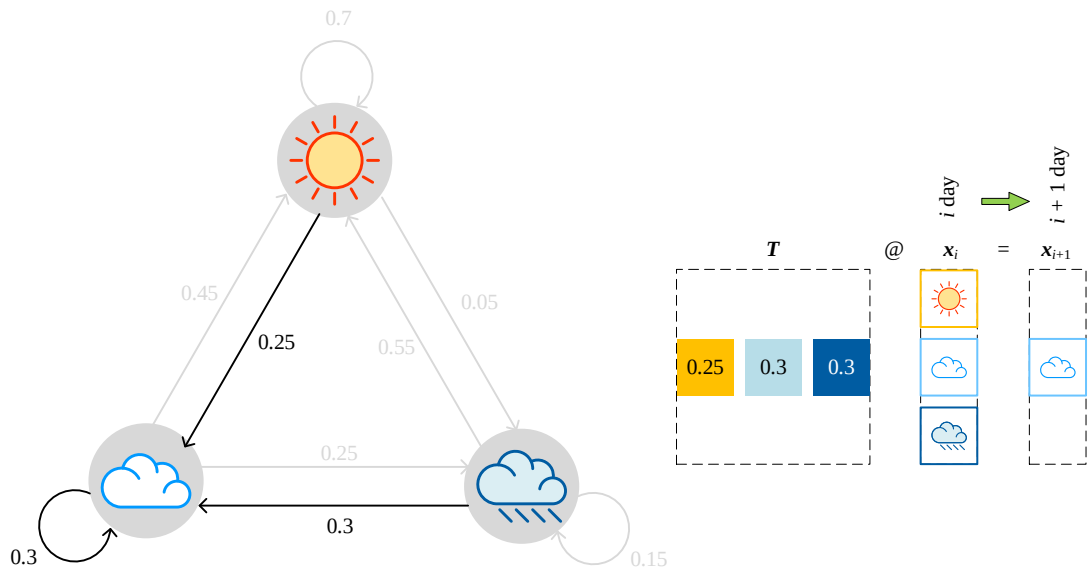


Figure 7. The Computation for Transitioning From the Three Current Weather States to a Cloudy Next Day

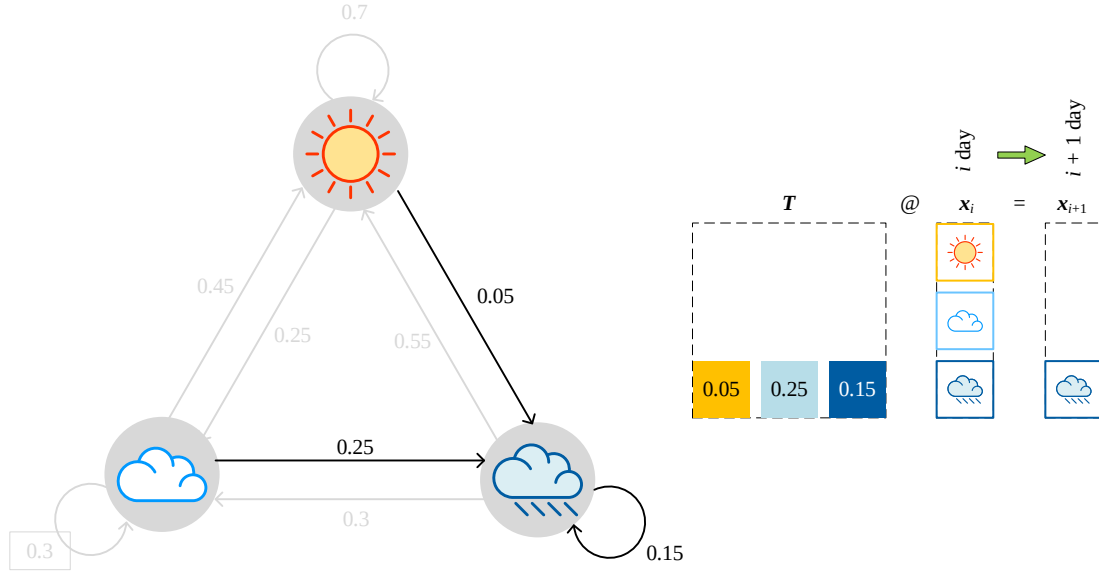


Figure 8. The Computation for Transitioning From the Three Current Weather States to a Rainy Next Day

Steady State

If the current weather is sunny, after 1 day, the weather's state vector is

$$x_{i+1} = T @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.25 \\ 0.05 \end{bmatrix}$$

After 2 days, the weather state is

$$x_{i+2} = T @ x_{i+1} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0.7 \\ 0.25 \\ 0.05 \end{bmatrix} = \begin{bmatrix} 0.63 \\ 0.265 \\ 0.105 \end{bmatrix}$$

After 3 days, the weather state is

$$x_{i+3} = T @ x_{i+2} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0.63 \\ 0.265 \\ 0.105 \end{bmatrix} = \begin{bmatrix} 0.618 \\ 0.2685 \\ 0.1135 \end{bmatrix}$$

After 4 days, the weather state is

$$x_{i+4} = T @ x_{i+3} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0.618 \\ 0.2685 \\ 0.1135 \end{bmatrix} = \begin{bmatrix} 0.61585 \\ 0.2691 \\ 0.11505 \end{bmatrix}$$

After 4 days, the weather state is

$$x_{i+5} = T @ x_{i+4} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix} @ \begin{bmatrix} 0.61585 \\ 0.2691 \\ 0.11505 \end{bmatrix} = \begin{bmatrix} 0.6154675 \\ 0.2692075 \\ 0.115325 \end{bmatrix}$$

As shown in Figure 9, after several days, the weather state vector x appears to converge to a stable value, that is

$$\mathbf{x} = T\mathbf{x}$$

The equation above tells us that the transition matrix has an eigenvalue of 1, and the corresponding eigenvector is the $\mathbf{x}$ in the equation above — a stable state. We call this $\mathbf{x}$ the steady state (steady state, stationary probability vector).

? Please analyze Figure 10 and Figure 11 on your own.

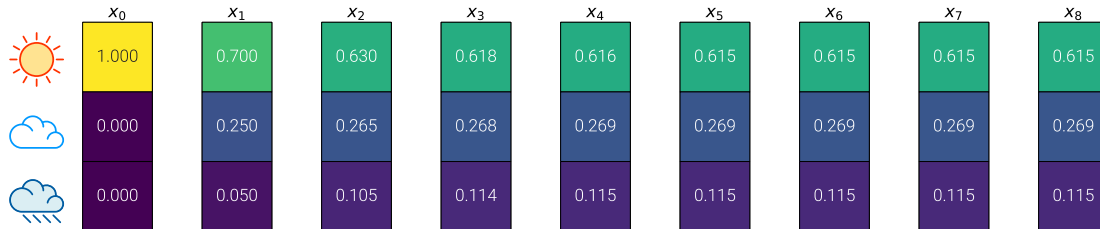


Figure 9. The Steady State Obtained From Sunny Weather via the Transition Matrix

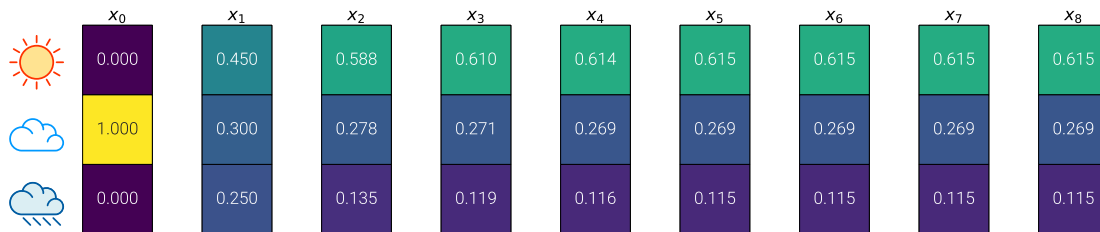


Figure 10. The Steady State Obtained From Cloudy Weather via the Transition Matrix

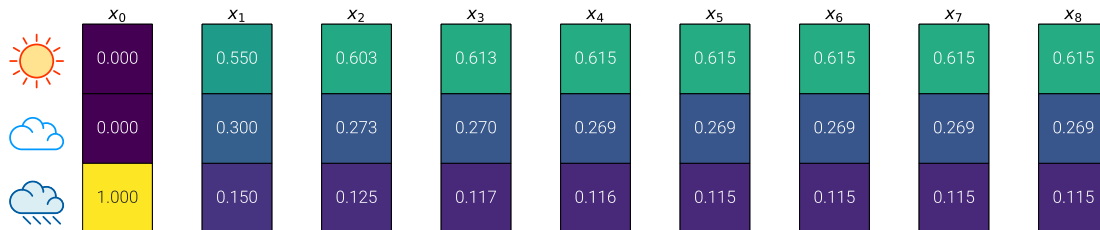


Figure 11. The Steady State Obtained From Rainy Weather via the Transition Matrix

This closely resembles the ball-passing problem from the previous section — except the question shifts from counting routes to computing probabilities, and each node now has a self-loop.

Eigenvalue Decomposition

Performing eigenvalue decomposition on transition matrix T

$$T = \underbrace{\begin{bmatrix} -0.902 & -0.791 & -0.289 \\ -0.395 & 0.222 & -0.516 \\ -0.169 & 0.569 & 0.805 \end{bmatrix}}_V \underbrace{\begin{bmatrix} 1 & & \\ & 0.178 & \\ & & -0.028 \end{bmatrix}}_A \underbrace{\begin{bmatrix} -0.681 & -0.681 & -0.681 \\ -0.584 & 1.119 & 0.506 \\ 0.269 & -0.933 & 0.739 \end{bmatrix}}_{V^{-1}} = \begin{bmatrix} 0.7 & 0.45 & 0.55 \\ 0.25 & 0.3 & 0.3 \\ 0.05 & 0.25 & 0.15 \end{bmatrix}$$

The equation above keeps only three digits after the decimal point.

The eigenvector corresponding to eigenvalue 1 is

$$\mathbf{v}_1 = \begin{bmatrix} -0.902 \\ -0.395 \\ -0.169 \end{bmatrix}$$

Since the steady-state vector's elements must all be positive, and must sum to 1, adjusting the eigenvector above gives

$$\mathbf{x}_{\text{steady}} = \begin{bmatrix} 0.615 \\ 0.269 \\ 0.115 \end{bmatrix}$$

The result keeps only three decimal places.

⚠ Note that normalizing a vector to unit length is different from normalizing a vector to sum to 1. After normalizing to unit length, the vector's L2 norm is 1; after normalizing to sum to 1, the vector's elements sum to 1.

This means that regardless of today's probabilities of sun, clouds, or rain, after enough time has passed, the weather probabilities converge to a fixed ratio, no matter what today's weather actually is.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please compute the eigenvalue decomposition of the following transition matrix, and compute the steady-state vector.

$$\mathbf{A} = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$$

Q2. Given the different initial state vectors below, apply the linear transformation using Q1's transition matrix, and compute the state vector after iterating each of these states 10 times

▶ $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

▶ $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

▶ $\mathbf{x}_0 = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$

Q3. Please use NetworkX to construct and visualize the directed graph shown in Figure 1, and write code to compute its adjacency matrix and degree matrix.